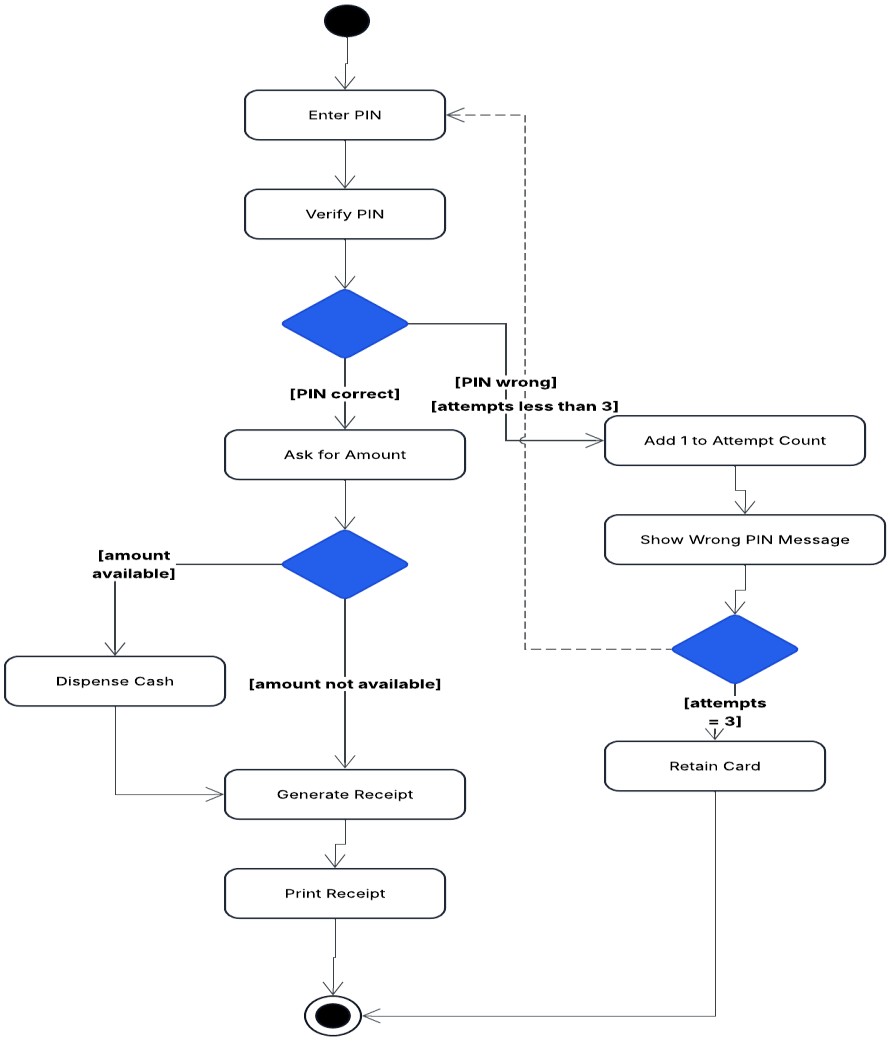
Southern New Hampshire University

DriverPass System Design Document

Madison Macachor

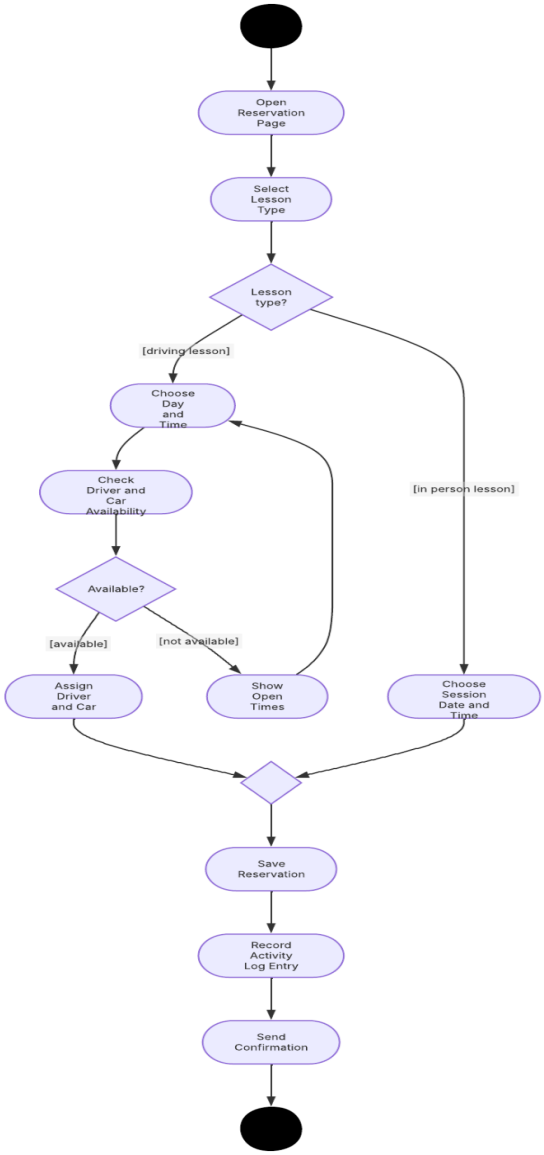
UML Diagrams

UML Use Case Diagram

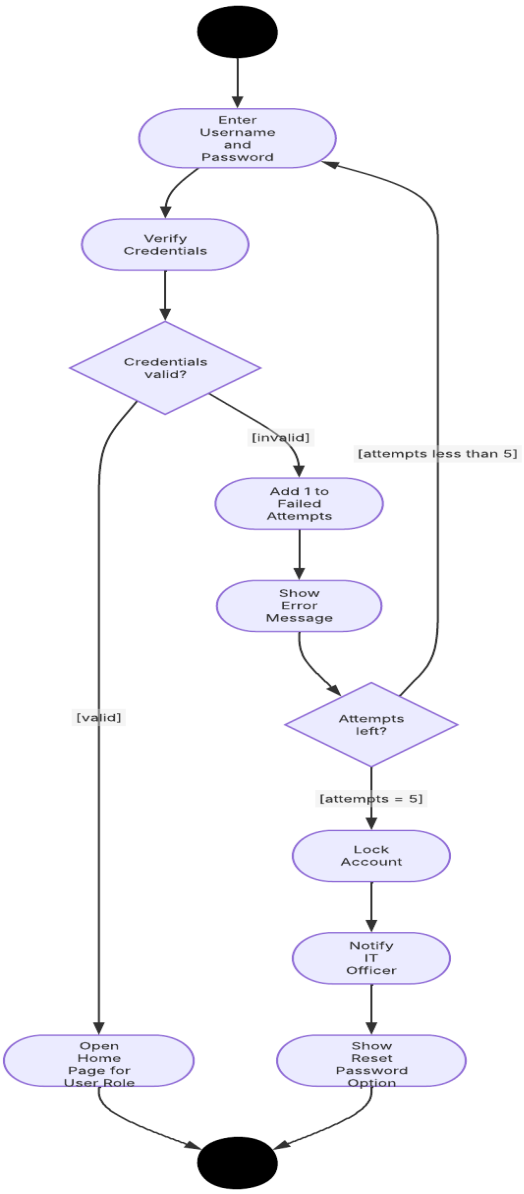


The use case diagram shows how each person and outside system uses DriverPass. Customers register, select a package, make, change, and cancel reservations, take online classes and practice tests, view their test progress, and contact DriverPass. The secretary makes, changes, and cancels reservations for customers who call or visit the office. The owner views reports, driver notes, and test progress. The IT officer manages accounts, resets passwords, blocks access, and enables or disables packages. Drivers enter lesson notes. The DMV system sends updates, and the payment service processes payments. Making, changing, and canceling a reservation each include recording an activity log entry, which gives the owner a record of who changed what.

UML Activity Diagrams



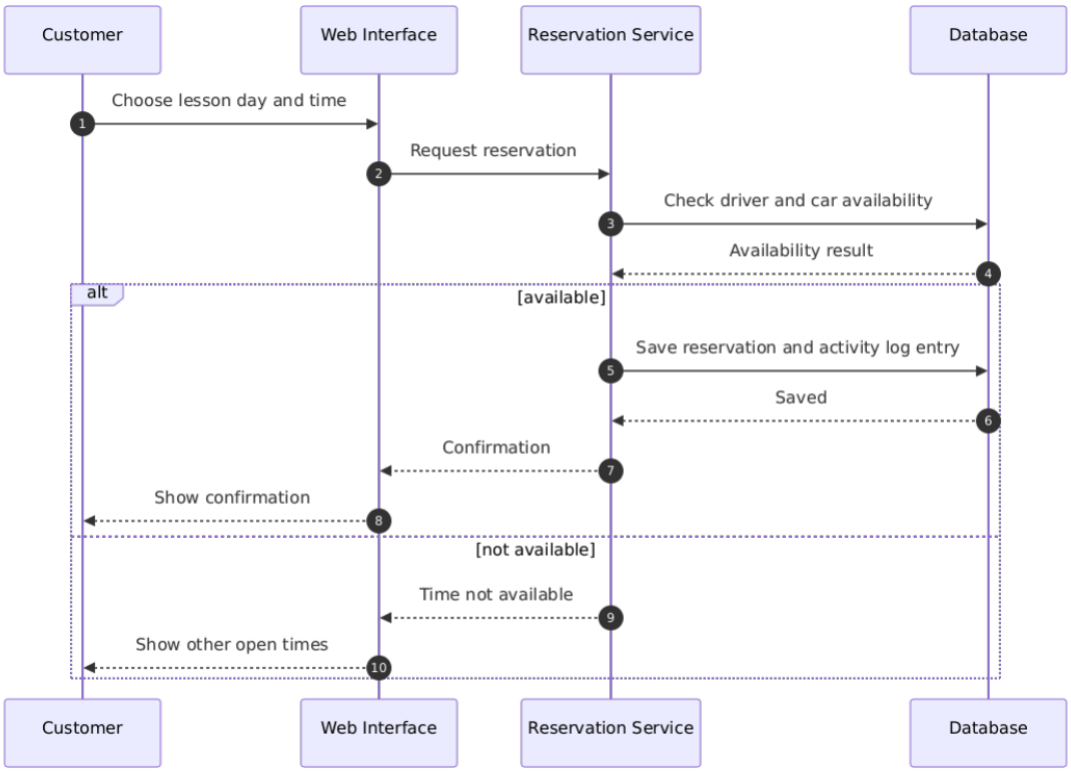
Make Reservation. The customer or the secretary starts by choosing a lesson type. For a driving lesson, the user picks a day and time, and the system checks driver and car availability. If both are free, the system assigns them. If not, it shows open times and the user picks again. For an in person lesson, the user picks a session date and time, and no driver or car is needed. Both paths then save the reservation, record an activity log entry, and send a confirmation. The activity log entry meets the owner's requirement to know who made, canceled, or changed each reservation.



Log In with Account Lockout. After the user enters a username and password, the system verifies them. Valid credentials open the home page for the user's role. Invalid credentials add one to the failed attempt count and show an error message, and the user can try again until the count reaches five. At

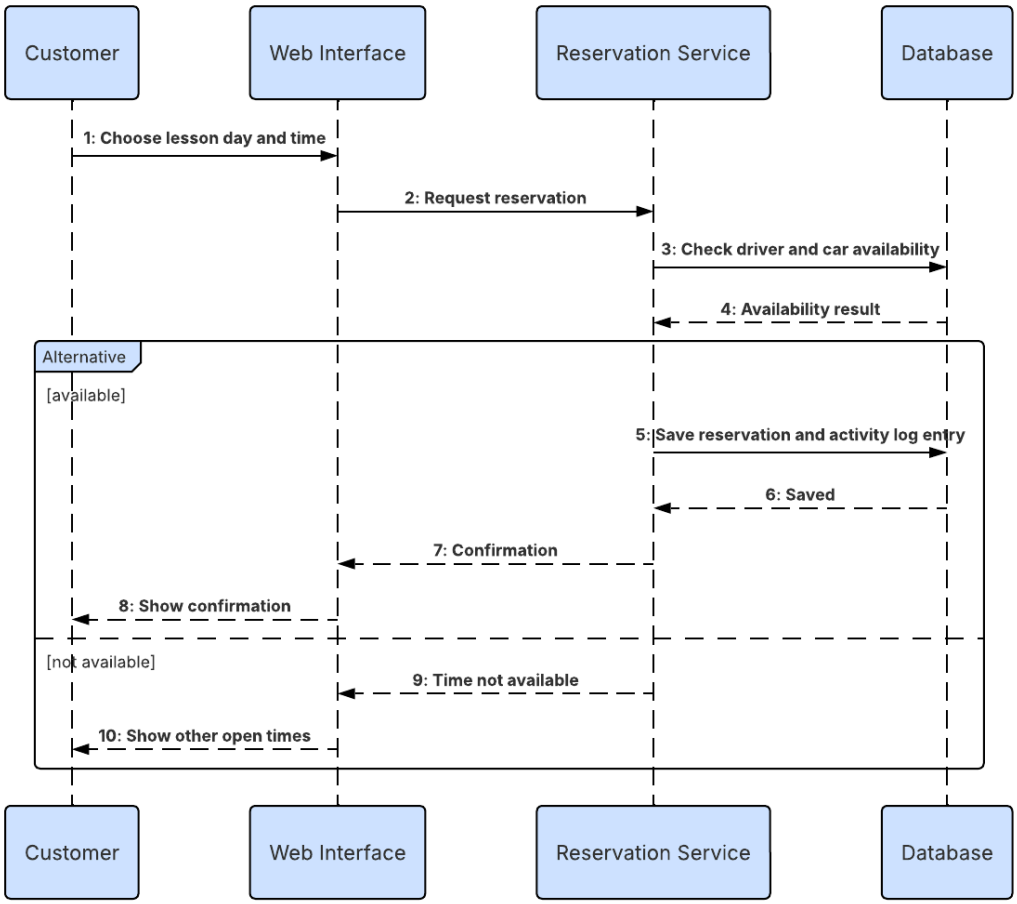
five, the system locks the account, notifies the IT officer, and shows a reset password option. This meets the security requirements for protecting against repeated login attempts and for letting users reset a forgotten password.

UML Sequence Diagram



This diagram follows a customer making a driving lesson reservation through the Web Interface, the Reservation Service, and the Database. The customer chooses a day and time, and the web interface passes the request to the reservation service. The service asks the database to check driver and car availability. If both are available, the service saves the reservation and an activity log entry, and the customer sees a confirmation. If not, the customer sees other open times. The alt box in the diagram shows these two outcomes.

UML Class Diagram



The class diagram shows the main information DriverPass stores. User holds the login and role information, and Customer and Driver add their own details. Secretary, Owner, and IT Officer have no extra attributes, so the role attribute covers them. Reservation is a parent class with two types. A DrivingLesson is linked to one driver and one car and has a two hour length. An InPersonLesson has a location and its own length, and it needs no driver or car, because the in person lesson on DMV rules and policies is not a driving session.

Package stores the hours and extras each package includes, along with an active flag so a package can be disabled without affecting current customers. PaymentMethod stores only a payment token and the last four card digits, because the payment service handles full card details. ActivityLogEntry records who did what to which reservation and when. TestResult stores time taken, score, and one of four statuses: not taken, in progress, failed, or passed. DMVUpdate stores the updates that change practice test content.

Technical Requirements

The requirements below come from the functional and nonfunctional requirements in the business requirements document. Speed targets and the account lockout limit are proposals, since the client didn't set numbers, and they should be confirmed with DriverPass.

Hardware

DriverPass won't buy or maintain servers. The system runs on virtual servers and a managed database service from a cloud provider, sized for a small business with 10 cars and room to grow. Staff and customers need only a computer, tablet, or phone with a current browser and an internet connection, so no special client hardware is required. The owner needs access to a printer to print activity reports. Payments are made online, so no card reader is needed.

Software

Web application. A responsive web application that works in current browsers on Windows, macOS, iOS, and Android. The front end uses standard HTML, CSS, and JavaScript. The back end uses a server side language and web framework chosen by the development team.

Database. A relational database, such as PostgreSQL or MySQL, stores the classes in the class diagram: users, customers, drivers, cars, packages, reservations, driver notes, test results, the activity log, and DMV updates. The reservation logic must stop a driver or car from being booked twice at the same time.

Authentication and access control. Users log in with a unique username and a password. Passwords are case sensitive and stored as hashes, never as plain text. Role based access control limits each user to the functions their role allows, and the IT officer has full control over accounts. After five failed logins the account locks and the IT officer is notified. Customers can reset a forgotten password through an emailed link. Sessions time out after a period of inactivity.

Encryption. All traffic uses HTTPS, and stored personal data is encrypted. The activity log can't be edited by regular users.

Payments. A third party payment service handles credit card payments, so DriverPass doesn't store full card numbers.

DMV connection. An interface receives DMV updates to rules, policies, and sample questions and notifies DriverPass when one arrives. This depends on what the DMV offers and needs to be confirmed.

Email. An email service sends reservation confirmations and password reset links. The registration form must collect an email address for this.

Reports. Reports can be downloaded in a format Excel opens, such as CSV or XLSX, and printed. Changes to data can only be made while online, so offline work is limited to downloaded reports.

Online class and tests. The online class and practice tests record each attempt with test name, time taken, score, and status.

Tools

Lucidchart for the UML diagrams and the schedule. Git for version control and an issue tracker for tasks and bugs. Automated testing tools for unit and integration tests, plus manual testing in each supported browser and on phones. Logging and monitoring tools that raise alerts for failed payments, failed DMV updates, server errors, and repeated failed logins. The IT officer receives these alerts.

Infrastructure

The cloud provider handles server security and automatic backups, as the client requested. The system needs separate development, testing, and production environments, so testing can happen before launch. It needs a domain name and a TLS certificate for secure connections. Maintenance and updates should be scheduled outside business hours. The proposed performance targets are pages that load in about three seconds and a reservation that saves in about two seconds under normal use. DriverPass staff need reliable internet at the office, since data can only be changed online.